TA5YY Propagation Monitor

Raspberry Pi 4 / Pi 5 • English Installation Guide • Version 1.1

Matching ZIP: TA5YY_Propagation_Monitor_RPi4_RPi5_TR_v1.1.zip

Developed by **TA5YY**. The dashboard displays a TA5YY signature; each operator configures their own station callsign.

1. Prerequisites

Use a 64-bit Raspberry Pi OS or Debian ARM64 installation with an internet connection, Docker Engine and Docker Compose. Do not perform a full operating-system upgrade solely for this application, especially on a Pi running other software such as qFT8.

```
clear
uname -m
docker --version
docker compose version
```

The architecture should report aarch64. If Docker and Compose display versions, skip to Step 3.

2. If Docker is not installed

Installing Docker changes system packages and services. Review the official instructions at https://docs.docker.com/engine/install/debian/ . The Debian 13 ARM64 test used the convenience script below. Inspect the downloaded script before running it; for managed systems, prefer Docker's official repository installation.

```
clear
cd ~
curl -fsSL https://get.docker.com -o get-docker.sh
less get-docker.sh
```

Press q to exit the viewer. Only if you accept the installation, run:

```
clear
cd ~
sudo sh get-docker.sh
sudo docker --version
sudo docker compose version
```

3. Extract the ZIP package

Copy the ZIP file to the Raspberry Pi desktop. The exact v1.1 filename below matches this guide.

```
clear
cd ~/Desktop
unzip TA5YY_Propagation_Monitor_RPi4_RPi5_TR_v1.1.zip
cd Propagation-Monitor-RPi
```

4. Install the application

Enter your own amateur-radio callsign, a 4/6/8-character Maidenhead locator, and an available web port between 1024 and 65535. The test used port 4434; the default is 8765.

```
clear
cd ~/Desktop/Propagation-Monitor-RPi
chmod +x kur.sh
sudo ./kur.sh
```

Important v1.1 fix: In v1, the regular user could not access the Docker socket and the dashboard could not write its SQLite database. The v1.1 installer configures the real user's UID/GID and the data directory permissions automatically. Do not use chmod 777.

5. Open and verify the dashboard

Open the Dashboard: http://PI_IP:PORT address printed by the installer from a browser on the same local network. The test IP 192.168.1.7 was specific to the test device; use the address shown on your own Pi.

```
clear
cd ~/Desktop/Propagation-Monitor-RPi
sudo docker compose ps
sudo docker compose logs --tail=20 dashboard
```

Both dashboard and mqtt services should show Up. Live reports may take time to arrive. If the page does not load, inspect the logs above.

6. Stop and restart

Run these commands from the extracted application directory.

```
clear
cd ~/Desktop/Propagation-Monitor-RPi
sudo docker compose down
# To start again:
sudo docker compose up -d
```

7. Security and limitations

The dashboard has no user authentication: do not expose its port directly to the internet. Map tiles require internet access. Historical PSK Reporter reports missed by the live MQTT feed are not automatically backfilled. The application does not decode RF signals itself; another reporting application must submit reception data to PSK Reporter. Personal records are stored in data/reports.db; do not distribute this database. Never reuse a previous operator's database when configuring another callsign.

Test status

On a Raspberry Pi 4 running Debian 13 ARM64, Docker and MQTT were confirmed working with v1. The dashboard opened after a manual SQLite directory permission fix. The v1.1 package incorporates fixes for these issues, but a full fresh end-to-end installation of v1.1 has not yet been verified.